

The University Of Azad Jammu & Kashmir,Muzaffarabad



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Course Title: DataBase Systems

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Lab 1- Database Laboratory

Part 01- Theory

1. Data and Information

Data

Data refers to raw facts and figures that do not have meaning by themselves.

Examples of Data:

- 85
- Ali
- 20
- Image file
- Audio recording

Information

Information is processed and organized data that provides meaning.

Example:

Data: Ali scored 85 marks in Database subject.

Another example:

Data: Temperature = 38°C

Information: Patient has fever.

2. Database and DBMS

Database

A database is an organized collection of data that can be easily accessed, managed and updated.

DBMS

DBMS stands for Database Management System. It is software used to store, retrieve and manage data efficiently.

Examples of DBMS:

- MySQL
- Oracle
- SQL Server
- PostgreSQL

3. Types of Data

Structured Data

Structured data is stored in rows and columns with a fixed format.

Example:

Roll No	Name	Marks
1	Ali	85
2	Sara	90

Structured data is stored in relational databases such as MySQL and Oracle.

Semi-Structured Data

Semi-structured data has some structure using tags or labels but does not follow a strict table format.

It is usually stored in JSON or XML format.

Example:

```
{ name: Ali, age: 20 }  
{ name: Sara, city: Lahore }
```

Unstructured Data

Unstructured data does not have any predefined format. It cannot be stored easily in rows and columns.

Examples:

- Images
- Videos
- Emails
- PDFs
- Audio files

Example: An X-ray image saved as /hospital/xray/patient10.png.

4. Relational and Non-Relational Databases

Relational Database

Relational databases store data in tables with a fixed schema and use SQL for queries.

Examples:

- MySQL
- Oracle
- SQL Server

Relational databases are mainly used for structured data.

Non-Relational Database

Non-relational databases store data in documents or files and have a flexible schema.

Examples:

- MongoDB
- Firebase
- Hadoop

Non-relational databases are used for semi-structured and unstructured data.

5. SQL

SQL stands for Structured Query Language. It is used to create, manage and retrieve data from databases.

SQL commands are divided into different categories:

DDL – Data Definition Language

DDL is called Data Definition Language because it defines the structure of database objects.

Commands:

- CREATE
- ALTER
- DROP
- TRUNCATE

DML – Data Manipulation Language

DML is called Data Manipulation Language because it manipulates the data stored in tables.

Commands:

- INSERT
- UPDATE
- DELETE
- SELECT

DCL – Data Control Language

DCL is called Data Control Language because it controls access and security of the database.

Commands:

- GRANT
- REVOKE

TCL – Transaction Control Language

TCL is called Transaction Control Language because it controls transactions.

Commands:

- COMMIT
- ROLLBACK
- SAVEPOINT

Part 02- Practical

Step 1: Installation of XAMPP



